

DS3040 Introduction to Deep Learning Lab

Assignment-7 , Date: Feb 23, 2025

Timing: 10:00 to 12:45 AM

Ungraded

Instructions

1. Submit one name.ipynb file containing all answers. The name should be [student_name]_assignment[number].ipynb
 2. Write the questions in separate text blocks before the answers.
 3. Outputs for all sub-questions should be given and the code should be executable.
 4. Write justifications for your choices where necessary.
 5. Ensure that all plots include clear labels and legends for better interpretation.
 6. Do not use AI or any websites except library documentations.
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Drive Link : https://drive.google.com/drive/folders/1mNr7rjZ1zuJmuJxqLLLp-HbPA4qKmo04?usp=drive_link

1. Pre-processing tasks:
 - a Load the given dataset using above drive link.
 - b Based on the given CSV file, split the dataset into train, valid, and test and print the distribution for each split.
 - c Show any five images, with the image ID as the title and the corresponding caption paragraph.
2. Image Feature Extraction
 - a Load the VGG16 model from keras and remove the last two layers from the architecture.
 - b Extract the features for each images and store them in variable with key image id and value extracted feature.
3. Sequence Model
 - a For image captions do the necessary preprocessing tasks (including lower-casing, removal of punctuation and stopwords etc.).
 - b Use tokenizer from tensorflow create a vocabulary and convert sequence of text into sequence of integers.
 - c Apply padding to the sequence of integers if applicable and use LSTM model to get the sequence features.
 - d Create the another model using addition or concatenation of image features and caption feature for image paragraph captioning task.

- e Train the model and plot the loss/accuracy curve.
 - f Generate caption for any 5 images from the test set.
- 4.
- a Experiment with the different pretrained image feature models, such as ResNet50, EfficientNet, etc., or you can create your own custom CNN model.
 - b Similarly, try different sequence models (like RNN, GRU, transformers, etc.) to get the better text feature representations.